Trivial Language Model

- Simplest option: empirical distribution over training sentences…

$$p(x_1 \ldots x_n) = \frac{c(x_1 \ldots x_n)}{N} \text{ for sentence } x = x_1 \ldots x_n$$

  - Problem: does not generalize (at all)
  - We need to assign non-zero probability to previously unseen sentences!

Unigram Model

- Generative process: pick a word, pick a word, pick a word … until you pick STOP

$$p(x_1 \ldots x_n) = \prod_{i=1}^{n} p(x_i)$$

  - Problem: these models disregard context completely